MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

1/ TYPE DOD-036

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pulsed, fixed nominal frequency of 13.325 GHz, integral magnet, 28 watts minimum, peak power output

ABSOLUTE RATINGS:

										Temperature	
Parameter:	Ef	tk	eb	ib	Ib	VSWR	tpc	trv	Alt	Backplate	Operating	Storage
Unit:	V	sec	v	ma	mAdc	---	µs	µs	ft	°C	°C	°C
Maximum:	6.62	---	1,200	150	75	---	6.25	---	75,000	+100	+85	+95
Minimum:	5.98	30	---	---	---	1.2	3.50	0.1	---	-65	-54	-62

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Marking:	See note 3
Mounting position:	Any	Magnet isolation:	2 inches (see note 3)
Mounting support:	Backplate (see figure 1)	Weight:	1.625 pounds (max)
Output coupling:	Mates with UG-541/U flange	Cooling:	Conduction

TEST CONDITIONS:

Parameter:	Ef(preheat)	Ef(operate)	tk	tpc	trv	Du	VSWR	Ib	T(backplate)
Unit:	V	V	sec	µs	µs	---	---	mAdc	°C
Tolerance:	±5%	±5%	Max	---	±0.035	---	Min	Max	±10
	6.3	---	30	5	0.095	0.5	1.2	70	60
	(Note 4)	(Note 4)	(Note 4)		(Note 5)	---	---	---	---

GENERAL:

First article sample inspection - Required (see note 1)
Shelf life - 18 months (see note 2)
Service-life guarantee - 400 hours

1/ Replaces:
Microwave Associates, Inc. types MA264 and MA281.
Singer-General Precision, Inc. Drawing numbers G181-170-001 and G181-170-002.

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	First article sample inspection						
1011	Humidity	6		---	---	---	---
1032	Vibration, mechanical	8, 15		---	---	---	---
1042	Shock, specified pulse	19	Test condition A, except 15 G; no voltages applied	---	---	---	---
1047	Low-temperature operation	7, 15	T (backplate) = -55° ±10°C	MP	---	1.0	%
1047	High-temperature operation	7, 15	T (backplate) = +90° ±10°C	MP	---	1.0	%
4027	Temperature coefficient	15, 20	T = -55° to +85°C	ΔF/T	-0.20	0	MHz/°C
4028	Barometric pressure, reduced	15	T (backplate) = -55° to +100°C; 75,000 ft	---	---	---	---
4266	Direct-interelectrode capacitance	-		Cin	18	25	pF
	Quality conformance inspection, part 1	9					
1367	Magnet polarity	-	North seeking pole adjacent to heater leads	---	---	---	---
4289	Heater current, nonoperating	4	Ef = 6.3 V; tk = 30 (max)	If	---	1.25	A
4286	Heater current, operating	4	Ef preselected	If	---	1.25	A
4303	Heater warmup time	4	Ef = 6.3 V; tk = 30 (max)	---	---	---	---
4304	Pulse characteristics	10		tpc	3.50	6.25	μs
4223	Mechanical tuning range	- -	*Upper limit* Lower limit	F F	13,300 ---	--- 13,370	GHz GHz
4306	Pulse voltage	5,11	Riding on bias voltage	epy	180	220	v
1261	Anode bias voltage	11		Eb	595	605	v
4310	Frequency pulling figure	12		ΔF	---	20	MHz
4311	Frequency pushing figure	13	ib = 130 to 150 ma	ΔF	---	1.0	MHz
4315	Pulse stability	7		MP	---	1.0	%
4250	Power output (average)	-		Po	14	20	W
4308	Minor lobe ratio	21		---	6	---	dB
---	Spurious output	11,18, appendix A	epy = 180 v (max)	spo/F	---	-45	*dBm/3 MHz*
	Quality conformance inspection, part 2						
1031	Low-frequency vibration	14, 15	No voltages applied	---	---	---	---
1031	*High-frequency vibration*	14, 15	*No voltages applied*	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3						
---	Life test	16, 17	Group D; T (backplate) = 90° ±10°C; VSWR = 1.2:1 (min), cycled through λg every 15 minutes (approx)	t	400	---	hrs
---	Life-test end points:						
4223	Mechanical tuning range	-	Upper limit	F	13,300	---	GHz
			Lower limit	F	---	13,370	GHz
4250	Power output (average)	-		Po	12.5	20	W
4306	Pulse voltage	5,11	Riding on bias voltage	epy	180	220	v
4311	Frequency pushing figure	13	ib = 130 to 150 ma	ΔF	---	1.3	MHz
4315	Pulse stability	7	Any 5-minute period	MP	---	1.0	%

NOTES:

1. First article sample inspection shall conform to the requirements of appendix F of MIL-E-1 and shall consist of performing all tests specified on this tube specification sheet (TSS). In addition, first article sample approval shall include satisfactory demonstration of tube-to-system compatibility.

   Invitation for bids should provide that the preparing activity reserves the right to waive the requirements for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

2. The tubes tested and procured in accordance with this TSS shall be capable of shelf life for a period of 18 months after date of shipment from the manufacturer's plant. Tubes placed in operation during *this shelf-life period* shall meet all requirements of this TSS. Tubes found to be defective within the shelf-life period shall be replaced by the manufacturer at no cost to the Government.

3. The "preheat" and "operate" voltages selected in accordance with note 4 and the words "CAUTION MAGNETS - KEEP 2 INCHES AWAY FROM IRON, STEEL, OR MAGNETIC MATERIALS" shall be marked on each tube at the location shown on figure 1.

4. Prior to the application of high voltage, the cathode shall be preheated to the required initial operating temperature by applying 6.3 V ±5 percent for 30 seconds maximum. Upon the application of anode voltage, the heater voltage shall be reduced to the "operate mode" by selecting one of the following voltages: 0, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, or 4.0. *The TUT shall then be capable of operating within the specified frequency* and power ranges within the time specified herein.

5. The trv shall be measured between the 20 percent and 85 percent points of the smooth peak amplitude. The voltage pulse shall exhibit no spikes and the slope of the rise shall not exceed 0.65 percent per nanosecond in the region above 50 percent.

6. The TUT output area normally covered by a choke flange in the system may be covered by an equivalent choke flange for this test.

7. Stability shall be measured in terms of the average number of output pulses missing, expressed as a percentage *of the number of input pulses applied during the period of observation.* The missing pulses (MP) due to any causes, are considered to be missing if the rf energy is less than 70 percent of the normal energy level. The stability shall be measured when VSWR of 1.2:1 minimum is introduced in the load and the phase is adjusted at the start of each measurement interval to produce maximum instability. The percentage of missing pulses shall not exceed the specified limit when measured in the last 3 minutes of a test interval not exceeding 6 minutes.

8. The tube shall be mounted in a resonance free jig and *vibrated with sinusoidal excitation in each of* three mutually perpendicular planes through the following amplitudes:

5 to 44 Hz	10 G (or 0.1 inch DA max)
44 to 500 Hz	10 G

   Cycling test: The frequency shall vary from 5 to 500 to 5 Hz with approximately logarithmic progression, and shall require approximately 30 minutes to traverse the range. This constitutes one cycle. The TUT shall be vibrated for one such cycle in each of the three planes.

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NOTES: -Continued

8. -Continued

Resonance test: Mechanical resonant frequencies of the tube shall be determined during the cycling test.

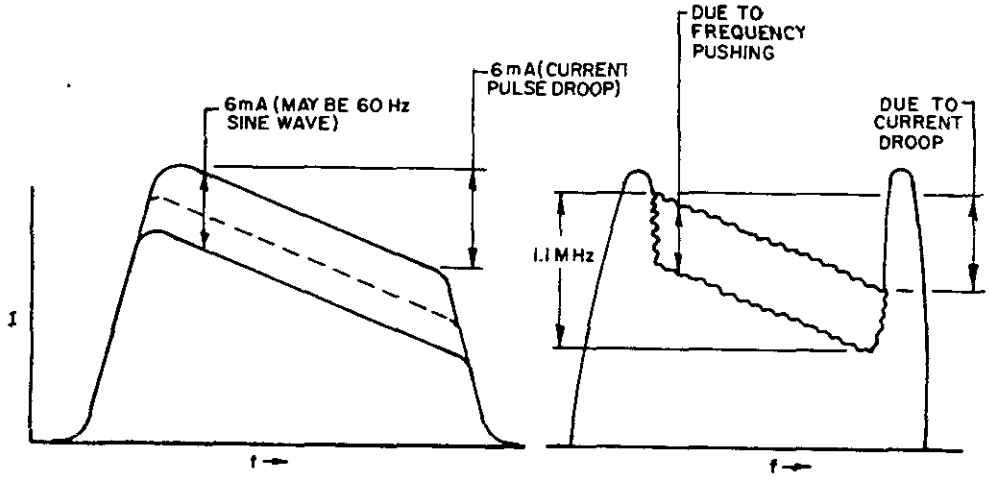
Vibration test schedule (times shown refer to one axis)

Type	Room temperature
Resonance	30 minutes
Cycling	30 minutes

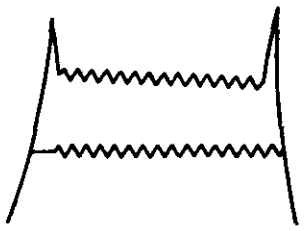
9. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

10. The pulse characteristics shall conform to the following requirements:

(a) Detected interferometer output depicting tolerance for ΔF over the peak current range 130 to 150 ma.



(b) Detected interferometer output depicting interpulse jitter.

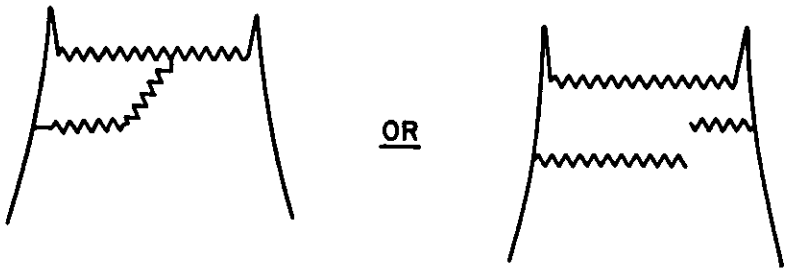


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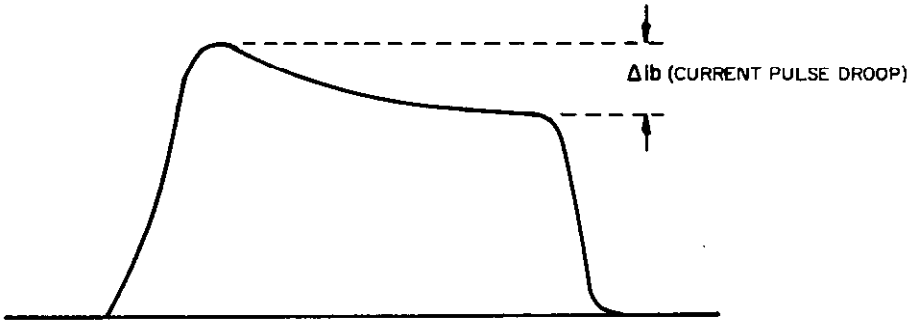
NOTES: -Continued

10. -Continued

(c) Detected interferometer output depicting intrapulse jitter.



(d) Current pulse droop characteristics as viewed on an oscilloscope using a 10:1 current probe.



NOTE: Δib shall be varied from 0 to 6 milliamperes during frequency deviation and jitter testing.

11. The anode voltage consists of +600 Vdc bias plus 180 to 220 volts pulse signals making the total applied potential 780 to 820 volts. See note 10(d) for additional pulse signal characteristics.

12. The frequency deviation (maximum frequency minus minimum frequency) shall not exceed the specified limits when a VSWR of 1.2 minimum is varied through all phases. A Singer-GPL modulator G7212-003, or equivalent is required for this test.

13. The frequency change shall be measured for a peak current change of 6 ma (130 to 136 ma, 137 to 143 ma, 144 to 150 ma) over the specified current range (a 60 Hz sine wave signal may be applied to obtain the 6 ma change in anode current). This requirement will include the frequency jitter. Measure the corresponding outputs in the microwave interferometer and determine the ΔF. To eliminate any amplitude effect present in this value, make two measurements of equal and opposite displacements from the phase null of the interferometer and use their average. Frequency jitter shall not contain any discrete components of frequency.

14. The TUT shall be vibrated for 1 minute in each of the three mutually perpendicular planes as shown on figure 1.

15. After the completion of the specified test, the TUT shall satisfy the power output, pulse voltage, and stability requirements of quality conformance inspection, part 1.

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NOTES: -Continued

16. Life test shall be conducted in accordance with the following cycle:

Condition	Ib	Ef	Duration
Stand by	0 mA	6.3 V	45 seconds
Operate	70 mA	Preselected *	2.0 hours
Off	0 mA	0	5.0 minutes

*See note 4

17. The standing wave introducer shall be moved during the test so that operation is obtained for load phase positions corresponding to maximum power output, minimum power output, maximum frequency, and minimum frequency. If automatically driven, the standing wave introducer shall be cycled continuously through a line length approximately one wave length long at a maximum rate of four cycles per hour; if manually moved, the standing wave introducer shall be through the four load phase positions corresponding to maximum power, minimum power, maximum frequency, and minimum frequency at least once during the specified life of the tube, spending approximately equal periods of time in each phase position.

18. Spurious output within ±1.5 MHz of the operating frequency shall be within the specified limits, when measured in accordance with the test procedure specified in appendix A of this TSS.

19. The TUT shall be subjected to five shocks of the specified peak amplitude and duration in each of three mutually perpendicular directions. Following impact tests, the TUT shall show no mechanical failure and shall meet the power output and pulse voltage requirements of quality conformance inspection, part 1.

20. Temperature measurements shall be made only after thermal equilibrium has been reached. The frequency shall be measured at the extremes of any 20°C temperature difference in the specified temperature range.

21. The radio frequency bandwidth and side lobes shall be within the limits specified when a VSWR of 1.2:1 minimum is introduced in the load and the phase is adjusted at the start of each measurement to produce maximum degradation. A satisfactory spectrum is one whose slope does not change sign more than once for power levels greater than 6 dB below its peak.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11
DSA - ES

User activities:
Army - ME, MU, WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2718)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
First article sample inspection		
G	---	1.250 (31.75)
H	---	2.125 (53.98)
N	---	.660 (16.76)
R	---	1.630 (41.40)
U	.473 (12.01)	.483 (12.27)
AG	.09 x .09 chamfer (2.29 x 2.29)	
AH	.765 (19.43)	.795 (20.19)
Quality conformance inspection, part 1 (see note 9)		
A	1.218 (30.94)	1.238 (31.45)
B	.450 (11.43)	.510 (12.95)
C	.262 (6.65)	.282 (7.16)
D	.073 (1.85)	.113 (2.87)
E	.063 (1.60)	.123 (3.12)
F	---	.775 (19.69)
P	.952 (24.18)	.960 (24.38)
K	.990 (25.15)	.998 (25.35)
V	.432 (10.97)	.442 (11.23)
Z	.248 (6.30)	.252 (6.40)
AA	.987 (25.07)	1.007 (25.58)
AC	.850 (21.59)	1.100 (27.94)
AE	---	.312 (7.92)
Quality conformance inspection, part 2		
J	1.485 (37.72)	1.515 (38.48)
L	---	.750 (19.05)
M	.485 (12.32)	.515 (13.08)
S	1.130 (28.70)	1.160 (29.46)
T	2.090 (53.09)	2.120 (53.85)
X	.460 (11.68)	.670 (17.02)
AD	1.720 (43.69)	1.910 (48.51)
AF	.560 (14.22)	.680 (17.27)
AJ	11.000 (279.40)	13.000 (330.20)
Reference dimensions		
AK	.310 x .630 (7.87) x (16.00)	
W	.190 (4.83)	
Y	.365 (9.27) Dia	
AB	.375 (9.53) Dia	

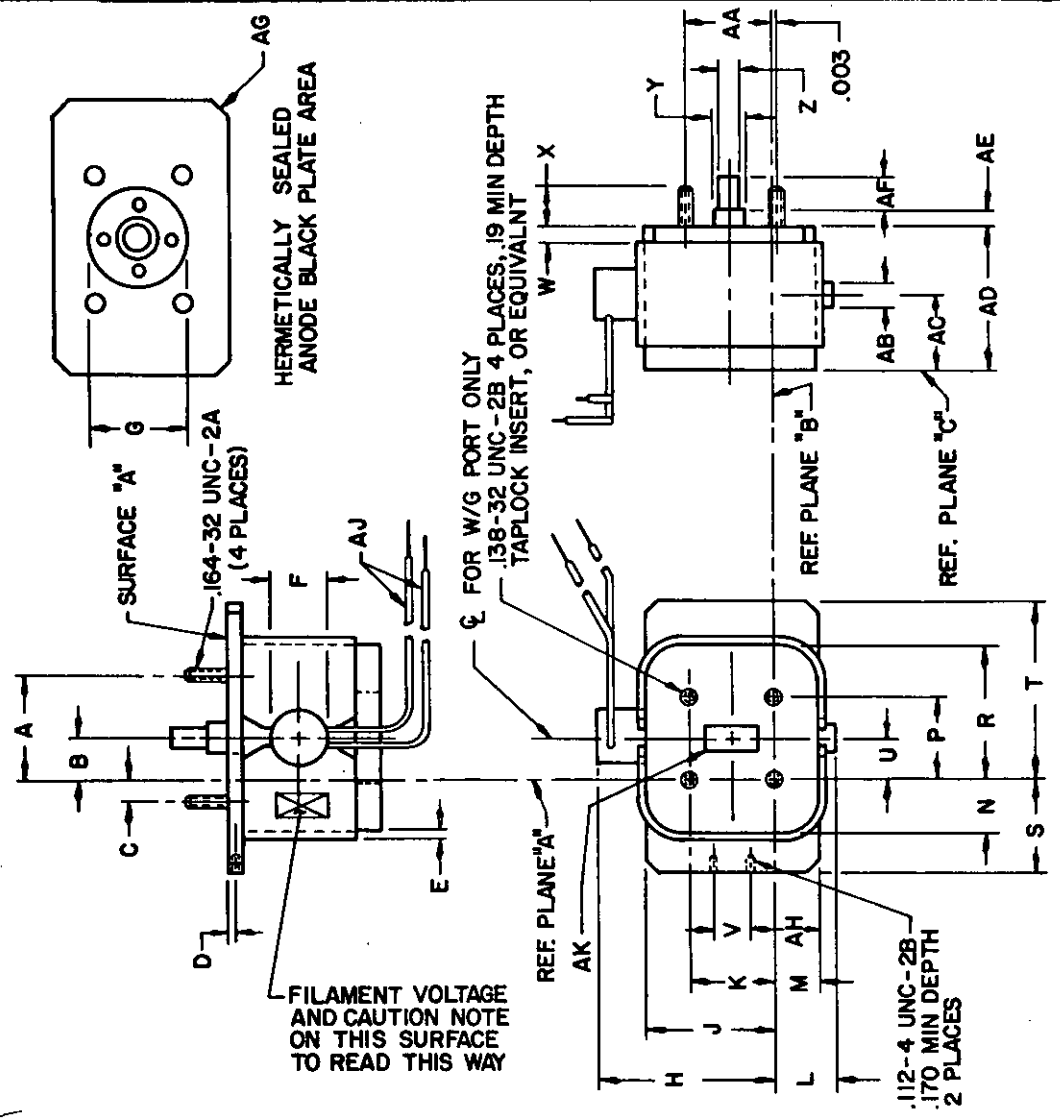


FIGURE 1. Outline drawing of electron tube type DOD-_____.

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APPENDIX A

TEST PROCEDURE

SPURIOUS OUTPUT

1. Purpose. The purpose of this test procedure is the measurement of spurious rf output, as a function of frequency.

2. Definitions. Spurious output signals are undesired outputs having a degree of coherence sufficient to distinguish them from white noise. Spurious output is the value of the ratio expressed in dBm/MHz between (1) the total spurious output emitted within a defined bandwidth and (2) the output of a known noise source within the same bandwidth.

3. Equipment. A microwave measurement system shall be assembled as shown on figure A-1. This system is intended to operate the TUT in the same manner that it is operated in the radar system and to accomplish a noise-power comparison between the TUT interpulse noise and a known noise source. The assembled test equipment shall include the following:

(a) The IF amplifier bandwidth shall be measurement bandwidth.

(b) The output shall be measured with a receiver that has a 3 MHz bandwidth.

(c) A high power load (VSWR 1.2 minimum) variable through all phases, a modulator, timer, and regulator which produces conditions equivalent to those of the radar system.

(d) A test load for the modulator having a dynamic impedance of 380 to 500 ohms @ 70 ma and a capacitance of 27 pF.

(e) Additional equipment.

High voltage power supply	-	Singer-Kearfott G7212-002, or equivalent.
High voltage regulator	-	Singer-Kearfott G7212-007, or equivalent.
Modulator	-	Singer-Kearfott G7212-003, or equivalent.
Directional coupler (10 dB)	-	Microwave Associates MA-706A, or equivalent.
Directional coupler (3 dB)	-	Hewlett-Packard P752A, or equivalent.
Switches (S1 and S3)	-	Singer-Kearfott G128-218-002, or equivalent.
	-	Microwave Associates MA83201Z10 (modified SKD units)
Switch S2	-	Singer-Kearfott G128-218-001, or equivalent.
Timer	-	Singer-Kearfott G7212-001, or equivalent.
Noise source	-	Microwave Semiconductor MC7133, or equivalent.
Local oscillator (LO)	-	Microwave Associates MA82K05, or equivalent.
Circulator (3 port)	-	Microwave Associates MA8M62, or equivalent.
Mixer	-	Microwave Associates MA69310, or equivalent.
IF amplifier	-	Singer-Kearfott G7298-008, or equivalent.
Digital voltmeter	-	Hewlett-Packard HP3440A, or equivalent.
Filament supply	-	LAMBDA LH118A, or equivalent.

4. Theory. The difficulty in performing the spurious output test is due to the wide dynamic range requirement in the test system, $(P_{\text{transmit dBm}} - P_{\text{noise dBm}}) = 80$ dBm. The modulator drives the TUT, whose power is conducted through the four-port circulator and a 10 dB directional coupler and a high-power attenuator AT-1 to switch S-1. Switches S1 and S2 are gated in the "STOP" mode using a drive pulse obtained from the timer during the transmit half-cycle. The coupler is optional if a high power load is used on port 3 of the 4-port circulator.

Each switch presents a 55 dB short (approximately) to the rf power delivered through the circulator, coupler, and attenuator, as well as providing a 110 dB isolation from the receiver. Energy reflected from the high reflection coefficient and seen at the input to switch S1 is attenuated by an additional 20 dB (second propagation through the 10 dB pad and coupler), and is subsequently dissipated in the termination connected at port 3 of the circulator. Leakage energy appearing at the receiver input terminals is in the order of -80 to -90 dBm.

During the receiver half cycle, switches S1 and S2 are forward biased (pass state), and the interpulse noise generated by the TUT is permitted to propagate to the receiver input terminal (attenuated by the line losses from the TUT to the mixer with AT-1 = X dB, and AT2 set for minimum). Receiver tuning is achieved during the transmit cycle by manual adjustment of the local oscillator.

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5. Procedure. The noise measurement is performed by injecting noise power into the receiver through switch S3 and directional coupler DC1, with the TUT "OFF". Switch S3 must be gated by either switch S1 or switch S2 drive signals in order to modulate the reference noise at 50 percent duty cycle (in the same manner that the interpulse noise is gated by the system modulator). This reference noise signal establishes s level on the IF output voltmeter. The noise source and S3 are disabled and the TUT and switches S1 and S2 are reactivated as described above. Attenuator AT2 is adjusted such that the reference level at the IF output is obtained. Since the gating of both the reference noise and the TUT noise is identical and since the noise source output level is known (KTΔB + dB), the TUT interpulse noise, P_m, may be calculated by the following equation:

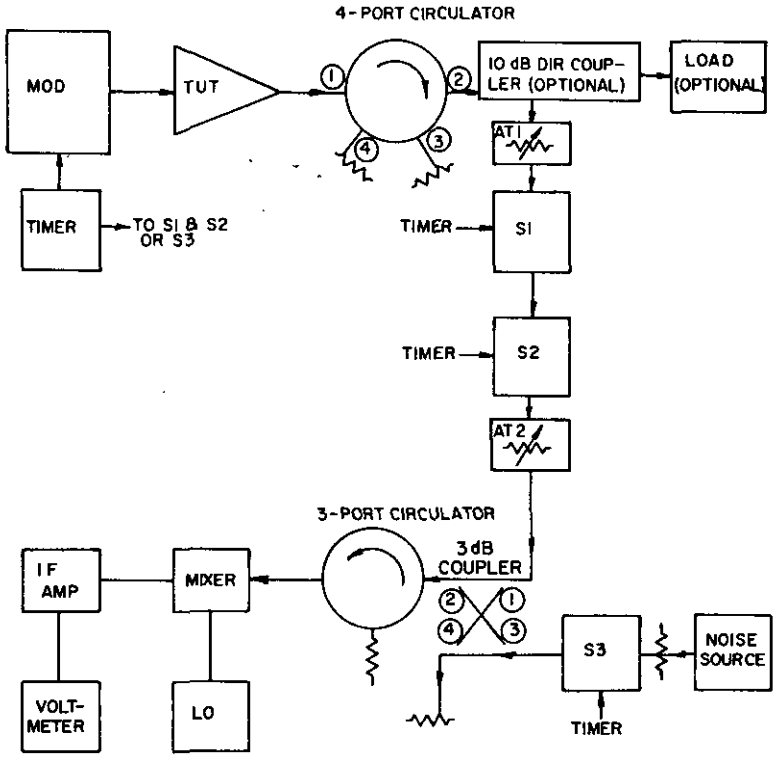
$$* P_{M\ dBm} = P_{Noise\ dBm} - (LS3 + LDC1_{3-2}) + (L_{line} + L_{Circ\ 1-2} + 10\ dB + LAT1 + LS1 + LS2 + AT2) + LDC1_{1-2}$$

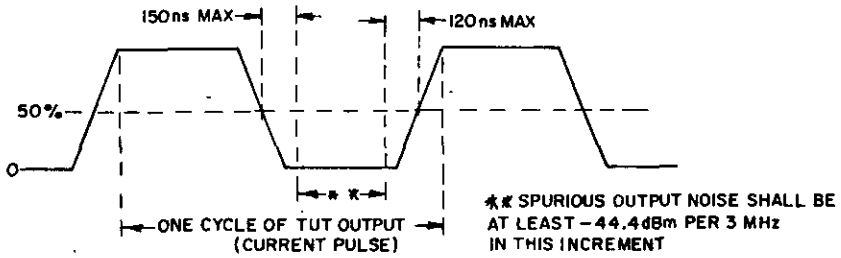
Where:

P_{noise} = KTΔB + 30 dB
KTΔB ≈ -109.23 dB (where B = 3 MHz)
KT = -114 dBm for 1 MHz of IF bandwidth
ΔB = IF bandwidth
LS3 = Pass-state insertion loss of S3
$LDC1_{3-2}$ = Directional coupler loss from port 3 to port 2
L_{line} = Line losses (i.e., waveguide necessary to make set up) from magnetron output flange to input flange of 3-port circulator
10 dB fixed = 10 dB directional coupler
LAT1 = loss in dB of AT1
LS1 = Pass state loss of S1
LS2 = Pass state loss of S2
$LDC1_{1-2}$ = Loss of directional coupler DC1 from port 1 to port 2 (3 dB coupler)
$L_{Circ\ 1-2}$ = Loss of 4-port circulator from port 1 to port 2

* All terms expressed in dB except where otherwise noted.

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Period for measuring spurious noise

FIGURE A-1. Test setup for measuring spurious noise.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products *can be procured with a minimum amount of delay and at the least cost.* Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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NAVAL ELECTRONIC SYSTEMS COMMAND
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DOD 316

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